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EDITORIAL COMMENT

Navigating the Landscape of Reinterventions After TAVR

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The rapid evolution of transcatheter aortic valve replacement (TAVR) has transformed the treatment of aortic stenosis. However, with increasing use in younger patients with longer life expectancies, the field faces the challenge of valve durability and reintervention. In this issue of *JACC*, Durand et al,[1] in their analysis of data from the FRANCE 2 and FRANCE TAVI (French Transcatheter Aortic Valve Implantation) registries, provide a comprehensive evaluation of the incidence, predictors, and outcomes of TAVR-related reinterventions. Their findings, grounded in nearly 73,000 patients treated over more than a decade, offer critical insights into the evolving paradigm of lifetime management of aortic stenosis (**Figure 1**).

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REINTERVENTION: A RARE BUT RAPIDLY GROWING CONCERN

While reintervention remains rare but increasingly relevant, management of transcatheter aortic valve (TAV) failure is limited to 2 options: redo-TAVR or TAVR-explant. Data from the FRANCE 2 and FRANCE TAVI registries showed a cumulative reintervention incidence of 1.7% at 8 years, consistent with other registries.[2,3] However, growing concern centers on the rising number of reinterventions, driven in part by the expansion of TAVR to low-risk patients. In the United States, TAVR-explant has become the fastest-growing cardiac operation in the Society of Thoracic Surgeons (STS) database, with a staggering 4,235% increase in case volume since 2012.[4] A similar trend is reflected in redo-TAVR,[5] highlighting the growing demand to systematically evaluate and manage these patients.

In the current study, median time to redo-TAVR was 0.33 years and 0.64 years for TAVR explant, with 62.1% of all reinterventions occurring within the first year. In contrast, the EXPLANTORREDO-TAVR registry reported longer times to reintervention, with redo-TAVR at 45.7 months and TAVR-explant at 17.6 months.[6] The discrepancy likely reflects differences in study population, as the EXPLANTORREDO-TAVR registry excluded emergency surgical conversions, in-hospital surgical interventions, and TAVR explants for endocarditis. In the current study, reintervention for endocarditis, although less common, accounted for 12% of all reinterventions, often requiring TAVR-explant as the definitive solution. Such differences underscore the need for standardized definitions of reintervention in the context of the mechanism of TAV failure to enable cross-study comparisons and inform clinical practice.

HIGH-STAKES DECISIONS: REDO-TAVR VS TAVR EXPLANT

Selecting the optimal strategy for managing failed TAVs is one of the most critical decisions facing heart teams today. Factors such as mechanism of index TAV failure, valve design, anatomic constraints, and patient comorbidities dictate the choice between redo-TAVR and TAVR explant.[7] Data from the EXPLANTORREDO-TAVR registry revealed that prosthesis-patient mismatch (PPM) was significantly more common in TAVR-explant cases (17.1% vs 0.5%), and unfavorable anatomy was present in 20% of explant cases, often precluding redo-TAVR. Conversely, short-frame valves and improved

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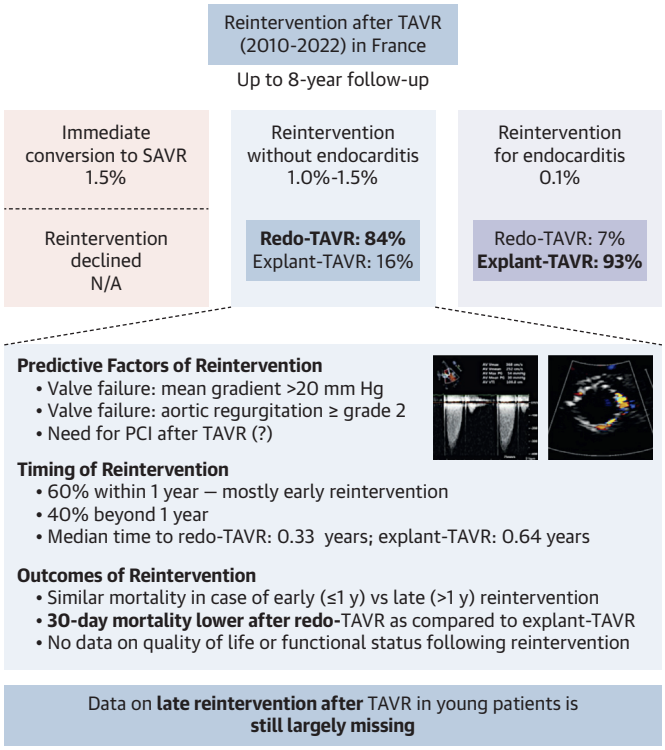
commissural alignment at the time of the index procedure facilitate redo-TAVR and reduce the need for TAVR explant. The lack of granular data on anatomic factors and the mechanism of TAV failure from the FRANCE TAVI registry limits our understanding of how these factors may have influenced reintervention strategy. Notably, TAVR-explant patients had higher residual gradients (mean 23 ± 5 mm Hg vs 15 ± 4 mm Hg) and more frequent moderate-to-severe aortic regurgitation (32% vs 18%) at index TAVR, often necessitating surgical intervention due to associated PPM and paravalvular regurgitation (PVR), respectively.

Redo-TAVR is generally favored due to its less invasive nature and more favorable short-term outcomes.[2,8] In the current study, redo-TAVR accounted for 84% of all reinterventions, reaffirming the feasibility of transcatheter approaches to manage TAV failure. However, the proportion of patients who were declined for any reintervention due to unfavorable patient or anatomic factors remains unknown. Redo-TAVR had lower early (7.5% vs 23.2%) and 6-year mortality (64.0% vs 76.2%) compared to TAVR explant, underscoring its procedural benefits. The EXPLANTORREDO-TAVR registry similarly reported a 4-fold increase in 30-day mortality for TAVR explant, but once patients survived the first 30 days, mortality rates remained similar between redo-TAVR and TAVR-explant.[6] In the current study, long-term mortality was comparable for early and late reintervention (57.3% vs 58.2% at 6 years), suggesting that procedural complexity and patient comorbidities, rather than timing, play a central role in determining outcomes.

FIGURE 1 Reintervention After TAVR: Insights From the FRANCE 2 and FRANCE TAVI Registries



PCI = percutaneous coronary intervention; SAVR = surgical aortic valve replacement; TAVR = transcatheter aortic valve replacement.

TAVR-EXPLANT: A LAST RESORT WITH COMPLEX CHALLENGES

Surgical explantation, often a last-resort option when redo-TAVR is not feasible, requires greater technical complexity, and is often performed in higher-risk patients with advanced disease states, contributing to its elevated mortality. Despite younger patients being better equipped to withstand surgery, as reflected in this study where TAVR-explant patients were younger (median age 78 vs 82 years), early mortality after TAVR-explant remains significantly higher. This disparity is likely driven by procedural complexity, patient selection, and the added risk of concomitant cardiac procedures. These findings highlight the need for meticulous preoperative planning, careful anatomic assessment, and optimal device selection during the index procedure to minimize complications requiring surgical intervention. Equally important is the standardization of surgical techniques to improve TAVR-explant outcomes.[9] Given the complexity and risks, surgeon experience and procedural volume play a critical role in patient outcomes. The present study lacked data on these key factors, underscoring the need for future research to explore the influence of operator expertise and procedural training to optimize outcomes and identify modifiable risk factors for reintervention.

PREDICTORS OF REINTERVENTION: A ROADMAP FOR RISK MITIGATION

Mitigating reintervention risk requires a proactive approach, beginning with optimal procedural planning and valve selection at the index procedure. Key procedural factors, such as elevated residual gradients, PVR, and PPM, have been linked to early valve dysfunction and the need for reintervention.[7] The FRANCE TAVI registry identified a mean aortic

gradient >20 mm Hg after TAVR as the strongest predictor of reintervention, doubling the risk compared to lower gradients. Moderate-to-severe PVR and first- or second-generation valves were also associated with higher reintervention rates, emphasizing the importance of technological advancements in mitigating risks. Advanced valve platforms like the SAPIEN 3 Ultra RESILIA (Edwards Lifesciences) have demonstrated superior hemodynamic performance, with lower rates of PVR and improved effective orifice areas.[10] Optimizing valve frame expansion during index TAVR is equally important to prevent structural deformation and hypoattenuating leaflet thickening, which are associated with early valve dysfunction.[11] Given the growing complexity of redo-TAVR, individualized patient-specific planning is essential. Emerging tools like the Redo-TAV App (Minneapolis Heart Foundation) emphasize preprocedural planning based on implant depth, device design, and coronary anatomy, while incorporating structured computed tomography analysis to guide decision-making.[12] The Heart Valve Collaboratory (HVC) framework highlights the importance of standardized pathways for evaluating THV failure, advocating for multimodality imaging to enhance systematic procedural planning and guide post-TAVR follow up, as part of a lifetime management strategy.[13]

CHARTING THE FUTURE: KEY LESSONS AND RESEARCH DIRECTIONS

The study by Durand et al[1] provides a strong foundation for future investigations into reinterventions after TAVR, yet several critical questions remain. Long-term durability remains uncertain, as the study addresses short- and mid-term outcomes without fully elucidating the comparative longevity of redo-TAVR and TAVR explant. Optimizing patient selection for initial interventions is crucial, requiring the development of robust predictive models to better stratify reintervention risk and guide procedural planning at index TAVR. Innovations in valve technology aimed at reducing PPM and improving coronary access will also play a pivotal role. While the present study focused primarily on mortality, assessing quality of life and functional status, particularly in a younger population with longer life expectancies, is essential to provide a more comprehensive understanding of the tradeoffs among reintervention strategies.

Durand et al[1] have highlighted the need for a paradigm shift in how we approach initial and subsequent interventions. Their findings underscore the importance of integrating anatomic, procedural, and patient-specific factors into decision-making, ensuring that each step in a patient's journey maximizes survival and quality of life. As the field continues to evolve, the insights from the FRANCE registries will undoubtedly serve as a cornerstone for shaping the future of TAVR.

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Dr Tang has received speaker's honoraria and served as a physician proctor, consultant, advisory board member, TAVR publications committee member, RESTORE study steering committee member, APOLLO trial screening committee member, and IMPACT MR steering committee member for Medtronic; has received speaker's honoraria and served as a physician proctor, consultant, advisory board member and TRILUMINATE trial anatomic eligibility and publications committee member for Abbott Structural Heart; has served as an advisory board member for Boston Scientific; has served as a consultant and physician screening committee member for Shockwave Medical; has served as a consultant for Philips, Edwards Lifesciences, Peija Medical, and Shenqi Medical Technology; and has received speaker's honoraria from Siemens Healthineers. Dr De Backer has received institutional research grants and consulting fees from Abbott, Boston Scientific, and Medtronic. Dr Zaid has reported that he has no relationships relevant to the contents of this paper to disclose.

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